SIKSHA 'O' ANUSANDHAN

(Deemed to be University)

Faculty of Engineering & Technology (ITER)

Department of Computer Science and Engineering

Project Proposal Form

FINAL YEAR RESEARCH PROJECT-2026

SECTION:41 **CSE** **GROUP NO: 14**

PROJECT TITLE: Web Advertisement Optimization Using Reinforcement Learning

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PROJECT ABSTRACT:

Optimizing web advertisement selection is critical for maximizing user engagement and revenue in digital advertising platforms. However, traditional ad selection methods rely on random or rule-based strategies that fail to efficiently balance exploration of new advertisements with exploitation of known high-performing ones, leading to suboptimal click-through rates (CTR). This gap motivates the need for an intelligent, adaptive ad selection mechanism that can learn in real-time without prior training data.

This project proposes a Reinforcement Learning-based solution using the Upper Confidence Bound (UCB) algorithm to address the web ad selection problem. The system simulates how an intelligent ad server learns from user interactions and dynamically adjusts its strategy to maximize total rewards. The UCB algorithm is benchmarked against epsilon-greedy and random selection strategies to establish its superiority in real-time decision making.

TECHNICAL SPECIFICATIONS

Software:

1. Python 3.x for data processing and simulation.
2. NumPy and Pandas for data handling and mathematical operations.
3. Matplotlib for performance visualization and result analysis.
4. Math library for UCB confidence score computation.

Hardware:

1. Windows/Linux with Python 3.x environment.
2. Google Colab / Jupyter Notebook for development and execution.
3. Standard CPU-based system (no GPU required).
4. Cloud resources (Google Colab) for execution and testing.

Methodology:

1. **Dataset Preparation:** Utilize a synthetic CSV dataset of 10,000 user interactions across 10 advertisements (binary: 1 = clicked, 0 = not clicked).

2. **Algorithm Design & Selection:** Implement the UCB algorithm to compute confidence scores for each advertisement, selecting the highest-scoring ad at each of the 10,000 simulated rounds.

3. **Comparative Analysis:** Benchmark UCB performance against epsilon-greedy and random selection baseline strategies to validate superiority.

4. **Output & Validation:** Evaluate total rewards, convergence behavior, and ad selection frequency to measure effectiveness and research contribution.

(1) SOFTWARE, HARDWARE OR METHODS/ALGORITHMS SPECIFICATIONS:

Programming Language: Python 3.x

Platform: Google Colab / Jupyter Notebook

Libraries Used: NumPy, Pandas, Matplotlib, Math

Algorithm: Upper Confidence Bound (UCB) — Reinforcement Learning / Multi-Armed Bandit

Dataset: Synthetic CSV dataset (dataset.csv) — 10,000 user interactions × 10 advertisements (binary: 1 = clicked, 0 = not clicked)

(2) NAME, REG. NO AND SIGNATURE OF GROUP MEMBERS:

1) Aditya Mishra	Reg. No: 2241013120	Signature: Aditya Mishra
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(3) APPROVAL STATUS (To be filled in by the Section Coordinator of FRP):

Prativa Das		
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Project Supervisor	**Section Coordinator, FRP**	**FRP Coordinator**